

# Vectors in space and operations on them

The same three operations as in the plane, with one extra component — and the position vector that makes coordinates and vectors the same thing.

LESSON 4–5

2 × 40 MIN

GEOMETRY 11, §2

P1 · VECTORS

LESSON CLOCK

10'

20'

22'

20'

18'

10'

|                                  |        |
|----------------------------------|--------|
| What carries over from the plane | 10 min |
| Components and the basis         | 20 min |
| The three operations             | 22 min |
| Magnitude and unit vectors       | 20 min |
| Position vectors                 | 18 min |
| Homework                         | 10 min |

LEARNING OBJECTIVES

- Write a vector in component and in unit-vector form.
- Add, subtract and multiply a vector by a scalar.
- Find the magnitude of a vector and a unit vector in its direction.
- Use position vectors to express  $\overrightarrow{AB}$  as  $\mathbf{b} - \mathbf{a}$ .

TEXTBOOK

|           |                              |
|-----------|------------------------------|
| National  | pp. 17–28                    |
| Cambridge | Core & Extended, pp. 325–334 |
| Workbook  | Exercise 2.1                 |

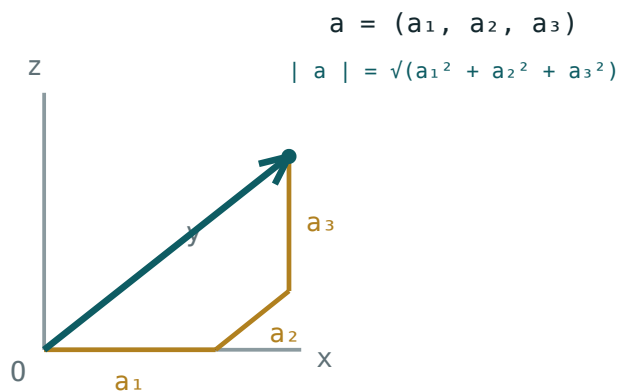
01

## Explanation

### One extra component, nothing else new

A vector in space is written in components or on the basis  $i, j, k$ :

$$\mathbf{a} = (a_1, a_2, a_3) = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$$



A vector as an arrow, and as the three steps along the axes that build it.

VECTOR OR POINT?

(3, 4, 5) as a **point** is a place. As a **vector** it is a displacement — three along, four across, five up — which may start anywhere. Two vectors are equal when their components agree, wherever they are drawn.

The three operations

| OPERATION       | RULE                          | EXAMPLE WITH $A = (1,2,3), B = (4,0,-1)$ |
|-----------------|-------------------------------|------------------------------------------|
| addition        | $(a_1+b_1, a_2+b_2, a_3+b_3)$ | $(5, 2, 2)$                              |
| subtraction     | $(a_1-b_1, a_2-b_2, a_3-b_3)$ | $(-3, 2, 4)$                             |
| scalar multiple | $(ka_1, ka_2, ka_3)$          | $3a = (3, 6, 9)$                         |

Geometrically nothing changes: addition is still the triangle rule, and  $ka$  is still  $a$  stretched by  $|k|$ , reversed if  $k$  is negative.

VECTORS ARE ADDED COMPONENTWISE

There is no “cross-multiplying” of components in addition. Each component minds its own axis.

Magnitude and unit vectors

$$|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

It is the distance formula from the origin, and it obeys the same rules as in the plane:  
 $|k\mathbf{a}| = |k| \cdot |\mathbf{a}|$ , and  $|\mathbf{a}| = 0$  only for the zero vector.

A **unit vector** has magnitude 1. To build one in the direction of  $a$ , divide by the magnitude:

$$\hat{a} = \frac{a}{|a|}$$

For  $a = (2, -1, 2)$ :  $|a| = 3$ , so  $\hat{a} = (\frac{2}{3}, -\frac{1}{3}, \frac{2}{3})$ .

**DIVIDE THE VECTOR, NOT THE COMPONENTS SEPARATELY IN YOUR HEAD**

Every component is divided by the **same** number  $|a|$ . Check by confirming the new magnitude is 1.

## Position vectors

**THE BRIDGE BETWEEN POINTS AND VECTORS**

The **position vector** of  $A$  is  $OA$ , written  $a$ , and its components are simply the coordinates of  $A$ . Then for any two points:

$$AB = b - a$$

“Head minus tail” — the destination’s position vector minus the start’s.

So  $A(1, 2, 3)$  to  $B(4, 6, 15)$  gives  $AB = (3, 4, 12)$ , of length 13 — the same answer as the distance formula, because it **is** the distance formula.

The midpoint of  $AB$  has position vector  $\frac{a+b}{2}$ , which is the midpoint formula written once instead of three times.

## 02 Worked examples

**EXAMPLE 1**  $a = (2, -1, 3)$ ,  $b = (1, 4, -2)$ . Find  $2a - 3b$  and its magnitude.

$$① \quad 2a = (4, -2, 6)$$

$$② \quad 3b = (3, 12, -6)$$

$$③ \quad 2a - 3b = (1, -14, 12)$$

$$④ \quad | \cdot | = \sqrt{1 + 196 + 144} = \sqrt{341} \approx 18.47$$

ANSWER

$$(1, -14, 12), \text{ magnitude } \sqrt{341} \approx 18.5$$

**EXAMPLE 2** Find a unit vector in the direction of  $a = (6, -2, 3)$ .

$$① \quad |a| = \sqrt{36 + 4 + 9} = 7$$

$$② \quad \hat{a} = \frac{1}{7}(6, -2, 3)$$

$$③ \quad \text{Check: } \frac{36 + 4 + 9}{49} = 1 \checkmark$$

ANSWER

$$\left(\frac{6}{7}, -\frac{2}{7}, \frac{3}{7}\right)$$

**EXAMPLE 3**  $A(3, 0, -1), B(1, 4, 5)$ . Find  $AB$ , its length, and the midpoint.

$$① \quad AB = b - a = (-2, 4, 6)$$

$$② \quad |AB| = \sqrt{4 + 16 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$③ \quad M = \frac{a + b}{2} = (2, 2, 2)$$

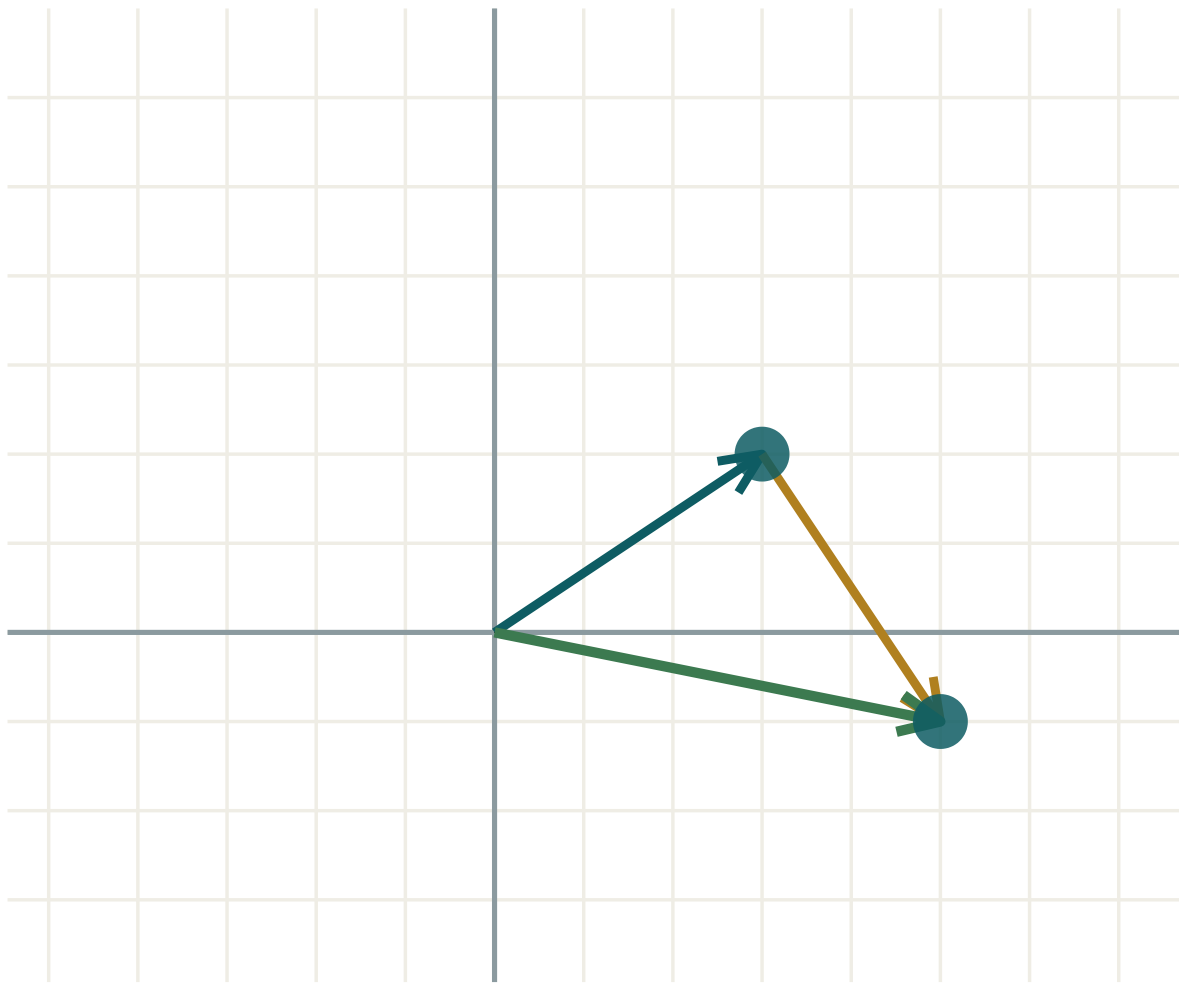
ANSWER

$$AB = (-2, 4, 6), 2\sqrt{14}, M(2, 2, 2)$$

### 03 Interactive model

Change the components and watch the arrow and its length move together.

## Vector arithmetic



$$a \ (3; \ 2)$$

$$b \ (2; \ -3)$$

$$a + b \ (5; \ -1)$$

$$|a| \ 3.61$$

$$|b| \ 3.61$$

$$|a + b| \ 5.1$$

### 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH                 | O'ZBEKCHA              | РУССКИЙ             |
|-------------------------|------------------------|---------------------|
| Vector                  | Vektor                 | Вектор              |
| Component               | Komponenta             | Компонента          |
| Position vector         | Radius-vektor          | Радиус-вектор       |
| Magnitude (modulus)     | Uzunlik (modul)        | Длина (модуль)      |
| Unit vector             | Birlik vektor          | Единичный вектор    |
| Zero vector             | Nol vektor             | Нулевой вектор      |
| Scalar multiplication   | Skalyarga ko'paytirish | Умножение на скаляр |
| Equal vectors           | Teng vektorlar         | Равные векторы      |
| Basis vectors $i, j, k$ | Bazis vektorlar        | Базисные векторы    |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

$|(3, 4, 12)|$  is:

19

13

12

$\sqrt{19}$

$AB$  in position vectors is:

$a + b$

$b - a$

$a - b$

$\frac{a + b}{2}$

A unit vector has magnitude:

0

1

$|a|$

any



$(1,2,3) + (4,0,-1)$  is:

$(5,2,2)$

$(4,0,-3)$

$(5,2,4)$

$(-3,2,4)$

$|3a|$  equals:

$|a|$

$3|a|$

$9|a|$

$|a|^3$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.  $(1,2,3) + (3,2,1)$

ANSWER  $(4,4,4)$

2.  $(5,0,2) - (1,3,2)$

ANSWER  $(4,-3,0)$

3.  $2(1,-2,3)$

ANSWER  $(2,-4,6)$

4.  $|(1,2,2)|$

ANSWER 3

5.  $|(0,3,4)|$

ANSWER 5

6.  $|(2,3,6)|$

ANSWER 7

7. Unit vector along  $(0,0,5)$

ANSWER  $(0,0,1)$

Medium · the standard question

7 PROBLEMS

1.  $a = (2, -1, 3)$ ,  $b = (1, 0, 4)$ ; find  $a + 2b$

ANSWER  $(4, -1, 11)$

2. Same; find  $3a - b$

ANSWER  $(5, -3, 5)$

3.  $|(6, -2, 3)|$

ANSWER 7

4. Unit vector along  $(6, -2, 3)$

ANSWER  $\frac{1}{7}(6, -2, 3)$

5.  $A(1, 2, 3)$ ,  $B(4, 6, 15)$ ; find  $AB$

ANSWER  $(3, 4, 12)$

6. Same; find  $|AB|$

ANSWER 13

7. Midpoint of  $A(1, 2, 3)$  and  $B(5, 6, 7)$

ANSWER  $(3, 4, 5)$

**Hard · stretch and reason**

7 PROBLEMS

1. Find  $k$  so  $|(k, 2, 2)| = 3$

ANSWER  $k = \pm 1$

2. Find a unit vector opposite to  $(1, 2, -2)$

ANSWER  $-\frac{1}{3}(1, 2, -2)$

3.  $a = (1, 2, 2)$ ,  $b = (2, -1, 2)$ ; find  $|a + b|$

ANSWER  $\sqrt{34} \approx 5.83$

4. Show  $|a + b| \leq |a| + |b|$  for those vectors

ANSWER  $5.83 \leq 6$

5. Find  $t$  so  $(1, t, 2)$  is parallel to  $(3, 6, 6)$

ANSWER  $t = 2$

6. Express  $(5, 4, 3)$  in terms of  $i, j, k$

ANSWER  $5i + 4j + 3k$

7.  $A(2, 1, 0)$ ,  $B(4, 5, 6)$ ; find the point  $P$  on  $AB$  with  $AP : PB = 1 : 3$

ANSWER  $(2.5, 2, 1.5)$

## 07 Homework

### Homework — 5 tasks

Check every unit vector by confirming its magnitude is 1.

- $a = (3, -1, 2)$ ,  $b = (0, 4, -2)$ . Find  $a + b$ ,  $a - b$  and  $2a + 3b$ .
- Find  $|a|$  and a unit vector along  $a$  for  $a = (2, 3, 6)$ .
- $A(0, 1, -2)$ ,  $B(6, 5, 4)$ . Find  $AB$ ,  $|AB|$  and the midpoint.
- Find  $k$  so that  $|(k, 4, 3)| = 13$ .

5. Explain in three sentences the difference between the point  $(2,3,4)$  and the vector  $(2,3,4)$ .